

Course : COMP8043041 – Machine Learning
Period : June 2023

Association Analysis

Session 05

Learning Outcomes

After the completion of this session, students are expected to achieve the following learning outcomes:

- LO2: Describe the characteristics of various Machine Learning algorithms and understand how each of them works, including the mathematical principles underlying the algorithms.
- LO3: Apply relevant Machine Learning algorithms according to individual cases/problems and perform evaluation.
- LO4: Analyze the results obtained from Machine Learning experiments from several perspectives.

Session Outcomes

By completing this session, you will be able to:

- Describe the concept of association analysis
- Describe the meaning of itemsets, support and confidence
- Summarize steps in Apriori algorithm
- Summarize steps in FP-Growth method
- Perform evaluation of association rules

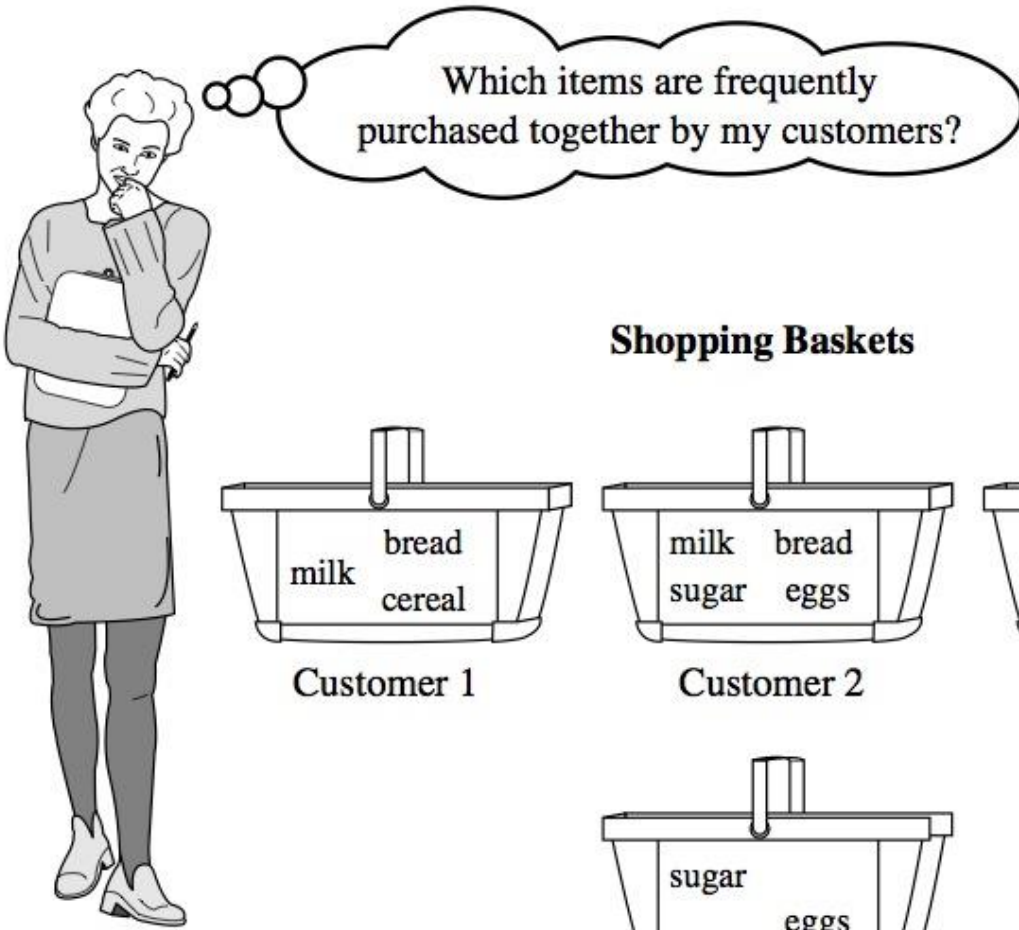
Topics

- Association Analysis
- Association Rules
- Itemsets, Support, and Confidence
- Apriori Algorithm
- Frequent Pattern Growth Methods
- Evaluation of Association Rules

What is Association Analysis?

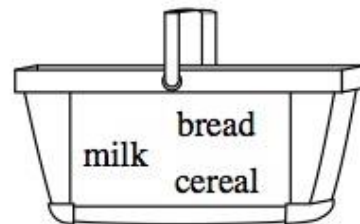
- Given a set of transactions, find association rules that will predict the occurrence of an item based on the occurrences of other items in the transaction.
- Applications:
 - basket data analysis
 - cross-marketing
 - catalog design
 - sale campaign analysis
 - Web log (click stream) analysis
 - DNA sequence analysis.

Illustration of Market Basket Analysis

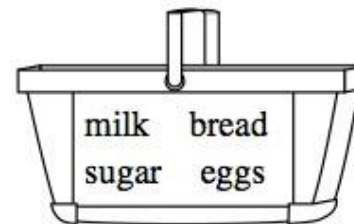


Which items are frequently purchased together by my customers?

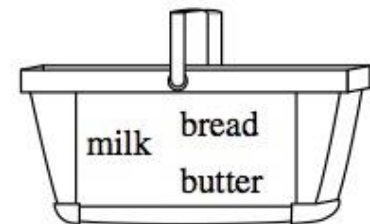
Shopping Baskets



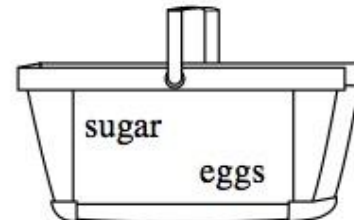
Customer 1



Customer 2



Customer 3



Customer n

Market Analyst

**People tend to buy
what others buy**



Business Use Case: Sales Transactions Analysis

- Huge amounts of purchase data are collected daily at the checkout counters of grocery stores.
- Finding sets of frequent items bought by customer can be used to support business-related decision making:
 - marketing promotions (cross-selling their products),
 - inventory management,
 - and customer relationship management.

Association Rules

- Given a set of transactions, find rules that will predict the occurrence of an item based on the occurrences of other items in the transaction.

Market-Basket transactions

<i>TID</i>	<i>Items</i>
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Example of Association Rules

$\{\text{Diaper}\} \rightarrow \{\text{Beer}\},$
 $\{\text{Milk, Bread}\} \rightarrow \{\text{Eggs, Coke}\},$
 $\{\text{Beer, Bread}\} \rightarrow \{\text{Milk}\},$

Implication means co-occurrence, not causality!

Frequent Itemset

- **Itemset**
 - A collection of one or more items
 - Example: {Milk, Bread, Diaper}
 - k-itemset
 - An itemset that contains k items
- **Support count (σ)**
 - Frequency of occurrence of an itemset
 - E.g. $\sigma(\{\text{Milk, Bread, Diaper}\}) = 2$
- **Support**
 - Fraction of transactions that contain an itemset
 - E.g. $s(\{\text{Milk, Bread, Diaper}\}) = 2/5$
- **Frequent Itemset**
 - An itemset whose support is greater than or equal to a *minsup* threshold. *Minsup*: minimum support.

Association Rule

□ Association Rule

- An implication expression of the form $X \rightarrow Y$, where X and Y are itemsets
- Example: $\{\text{Milk, Diaper}\} \rightarrow \{\text{Beer}\}$

<i>TID</i>	<i>Items</i>
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

□ Rule Evaluation Metrics

- Support (s)
 - ◆ Fraction of transactions that contain both X and Y
- Confidence (c)
 - ◆ Measures how often items in Y appear in transactions that contain X

Example: $\{\text{Milk, Diaper}\} \Rightarrow \text{Beer}$

$$s = \frac{\sigma(\text{Milk, Diaper, Beer})}{|T|} = \frac{2}{5} = 0.4$$

$$c = \frac{\sigma(\text{Milk, Diaper, Beer})}{\sigma(\text{Milk, Diaper})} = \frac{2}{3} = 0.67$$

Association Rule Mining Task

- Given a set of transactions T , the goal of association rule mining is to find all rules having:
 - support $\geq$ *minsup* threshold
 - confidence $\geq$ *minconf* threshold
- Brute-force approach:
 - List all possible association rules
 - Compute the **support** and **confidence** for each rule
 - Prune rules that fail the *minsup* and *minconf* thresholds

⇒ **Computationally prohibitive!**

Mining Association Rules

Example of Rules:

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

$\{\text{Milk, Diaper}\} \rightarrow \{\text{Beer}\} \text{ (s=0.4, c=0.67)}$
 $\{\text{Milk, Beer}\} \rightarrow \{\text{Diaper}\} \text{ (s=0.4, c=1.0)}$
 $\{\text{Diaper, Beer}\} \rightarrow \{\text{Milk}\} \text{ (s=0.4, c=0.67)}$
 $\{\text{Beer}\} \rightarrow \{\text{Milk, Diaper}\} \text{ (s=0.4, c=0.67)}$
 $\{\text{Diaper}\} \rightarrow \{\text{Milk, Beer}\} \text{ (s=0.4, c=0.5)}$
 $\{\text{Milk}\} \rightarrow \{\text{Diaper, Beer}\} \text{ (s=0.4, c=0.5)}$

Observations:

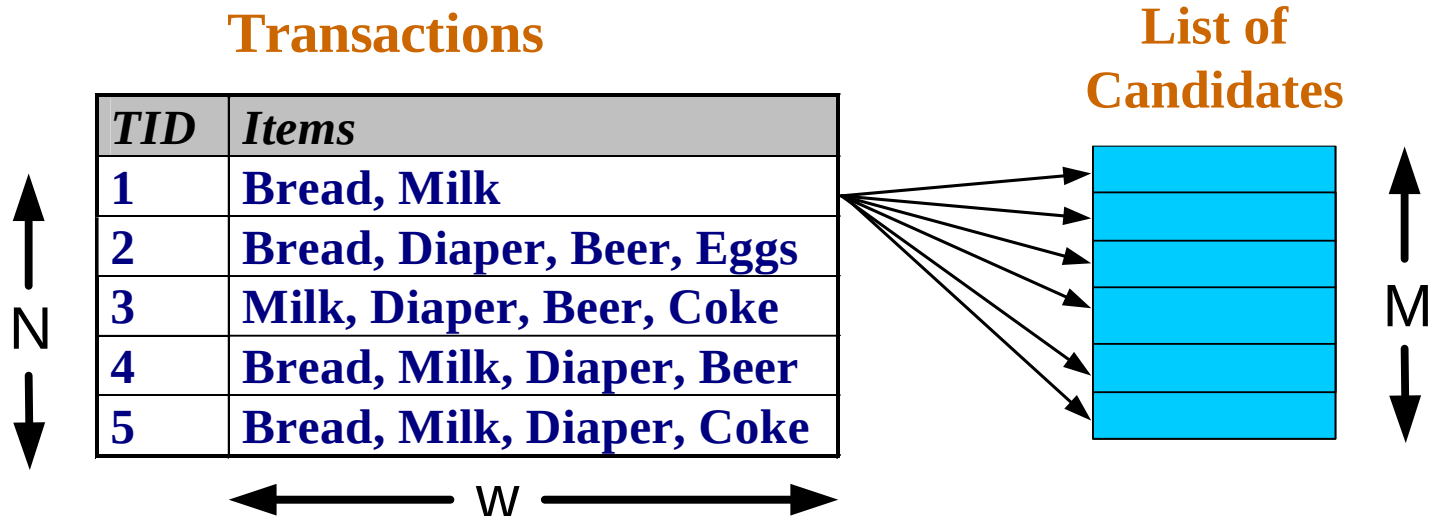
- All the above rules are binary partitions of the same itemset:
 $\{\text{Milk, Diaper, Beer}\}$
- Rules originating from the same itemset have identical support but can have different confidence.
- Thus, we may decouple the support and confidence requirements.

Mining Association Rules

- Two-step approach:
 1. Frequent Itemset Generation
 - Generate all itemsets whose support $\geq$ minsup.
 2. Rule Generation
 - Generate high confidence rules from each frequent itemset, where each rule is a binary partitioning of a frequent itemset.
- Frequent itemset generation is still computationally expensive.

Frequent Itemset Generation

- Brute-force approach:
 - Each itemset in the lattice is a **candidate** frequent itemset.
 - Count the support of each candidate by scanning the database.



- Match each transaction against every candidate
- Complexity $\sim O(NMw) \Rightarrow$ **Expensive since $M = 2^d$!!!**

Apriori Algorithm

Method:

- Let $k=1$
- Generate frequent itemsets of length 1.
- Repeat until no new frequent itemsets are identified:
 - Generate length $(k+1)$ candidate itemsets from length k frequent itemsets.
 - Prune candidate itemsets containing subsets of length k that are infrequent.
 - Count the support of each candidate by scanning the DB.
 - Eliminate candidates that are infrequent, leaving only those that are frequent.

Illustrating Apriori Principle

Item	Count
Bread	4
Coke	2
Milk	4
Beer	3
Diaper	4
Eggs	1

Items (1-itemsets)



Itemset	Count
{Bread,Milk}	3
{Bread,Beer}	2
{Bread,Diaper}	3
{Milk,Beer}	2
{Milk,Diaper}	3
{Beer,Diaper}	3

Pairs (2-itemsets)

(No need to generate candidates involving Coke or Eggs)



Triplets (3-itemsets)

Itemset	Count
{Bread,Milk,Diaper}	3



Minimum Support = 3

If every subset is considered,
 ${}^6C_1 + {}^6C_2 + {}^6C_3 = 41$
 With support-based pruning,
 $6 + 6 + 1 = 13$

Factors Affecting Complexity

- Choice of minimum support threshold
 - Lowering support threshold results in more frequent itemsets.
 - This may increase number of candidates and max length of frequent itemsets.
- Dimensionality (number of items) of the data set
 - More space is needed to store support count of each item.
 - If number of frequent items also increases, both computation and I/O costs may also increase.
- Size of database
 - Since Apriori makes multiple passes, runtime of algorithm may increase with number of transactions.
- Average transaction width
 - Transaction width increases with denser datasets.
 - This may increase max length of frequent itemsets and traversals of hash tree (number of subsets in a transaction increases with its width).

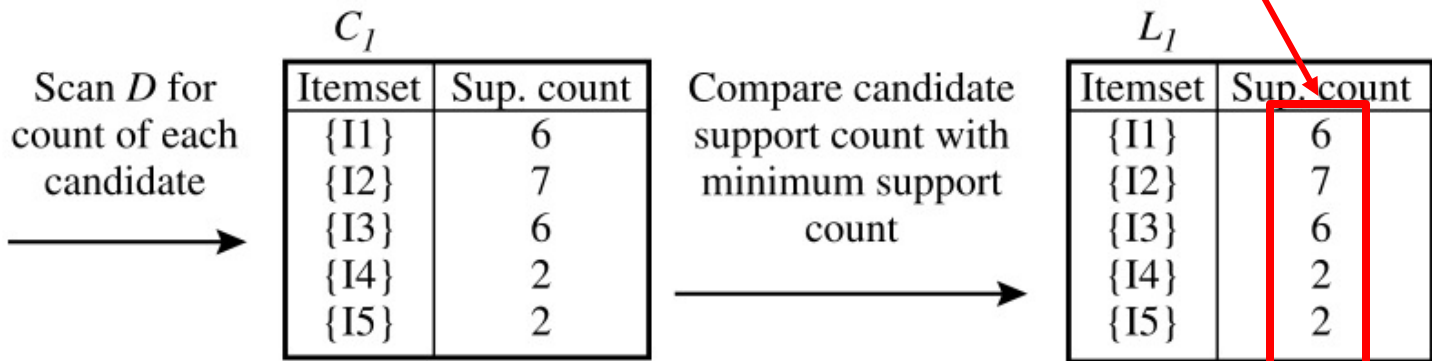
The Apriori Algorithm: Example

<i>TID</i>	<i>List of item IDs</i>
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3

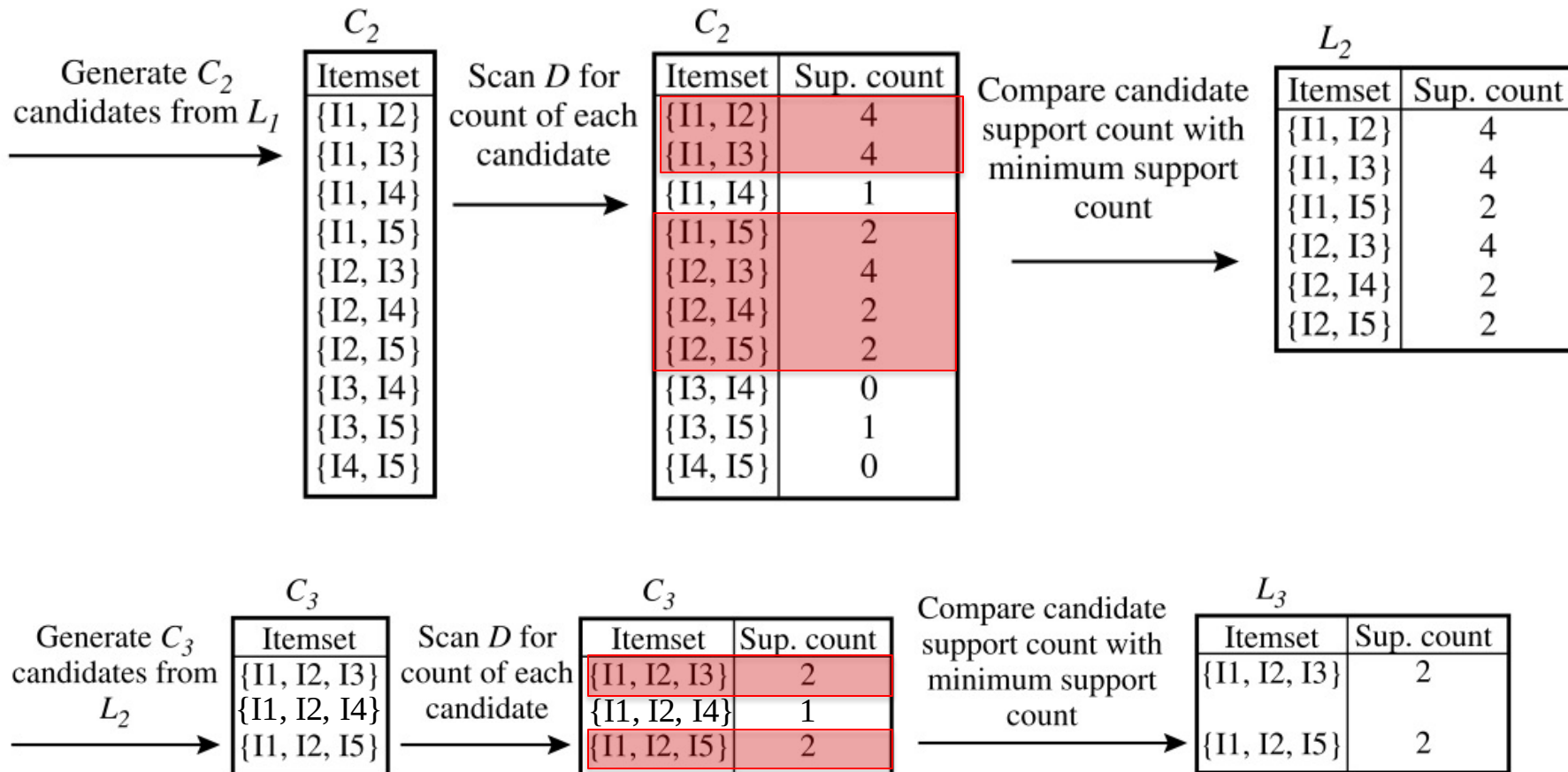
Find ***frequent itemsets*** with
minimum support count equals
2!

minSupCount = 2

supCount ≤ minSupCount



The Apriori Algorithm: Example (cont.)



The Apriori Algorithm: Example (cont.)

Generate C4
candidates from L3



L_4

Itemset	Sup. count
{I1, I2, I3, I5}	

Doesn't pass the threshold

DONE

Output:

The frequent itemsets that meet the requirement:

- {I1, I2, I3}
- {I1, I2, I5}
- {I1, I2}
- {I1, I3}
- {I1, I5}
- {I2, I3}
- {I2, I4}
- {I2, I5}
- {I1}
- {I2}
- {I3}
- {I4}
- {I5}

Generating Association Rules from Frequent Itemsets

- Association rules can be generated as follows:
 1. For each frequent itemset l , generate all nonempty subsets of l
 2. For every nonempty subset s of l , output the rule " $s \Rightarrow (l - s)$ " if $\text{support_count}(l) / \text{support_count}(s) \geq \text{min_conf}$
- Example:
 - Suppose the data contain the frequent itemset $l = \{I1, I2, I5\}$
 - The nonempty subsets of l are : $\{I1, I2\}$, $\{I1, I5\}$, $\{I2, I5\}$, $\{I1\}$, $\{I2\}$, and $\{I5\}$
 - The resulting association rules:

Rule	Confidence
$I1 \wedge I2 \rightarrow I5$	$2/4 = 50\%$
$I1 \wedge I5 \rightarrow I2$	$2/2 = 100\%$
$I2 \wedge I5 \rightarrow I1$	$2/2 = 100\%$
$I1 \rightarrow I2 \wedge I5$	$2/6 = 33\%$
$I2 \rightarrow I1 \wedge I5$	$2/7 = 29\%$
$I5 \rightarrow I1 \wedge I2$	$2/2 = 100\%$

Suppose $\text{min_conf} = 70\%$

FP Growth

<i>TID</i>	<i>List of item_IDs</i>
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3

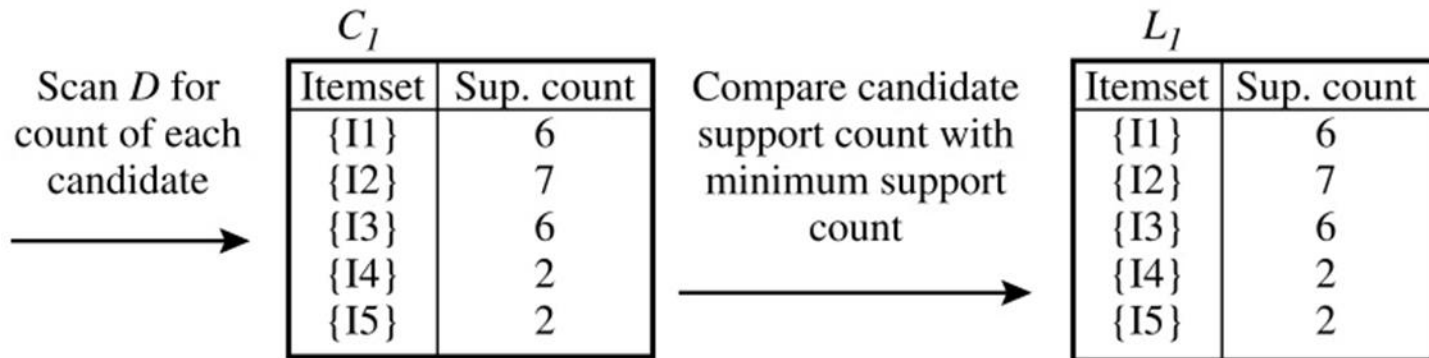
- Langkah 1 : Tentukan minimum support dan minimum confidence

Mis :

sup min : 2

Conf min : 2

- Langkah 2 : Generate frequent item set

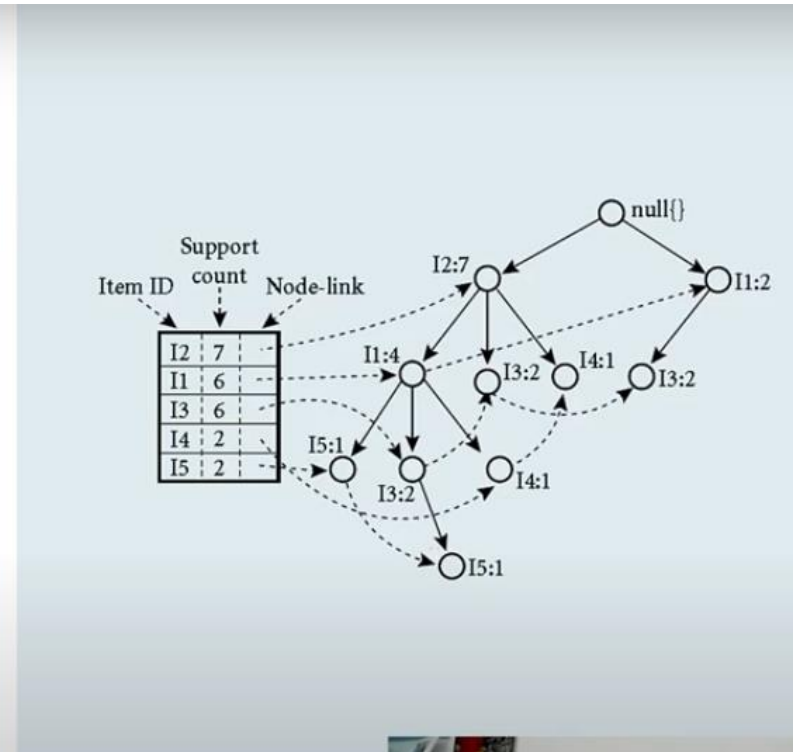
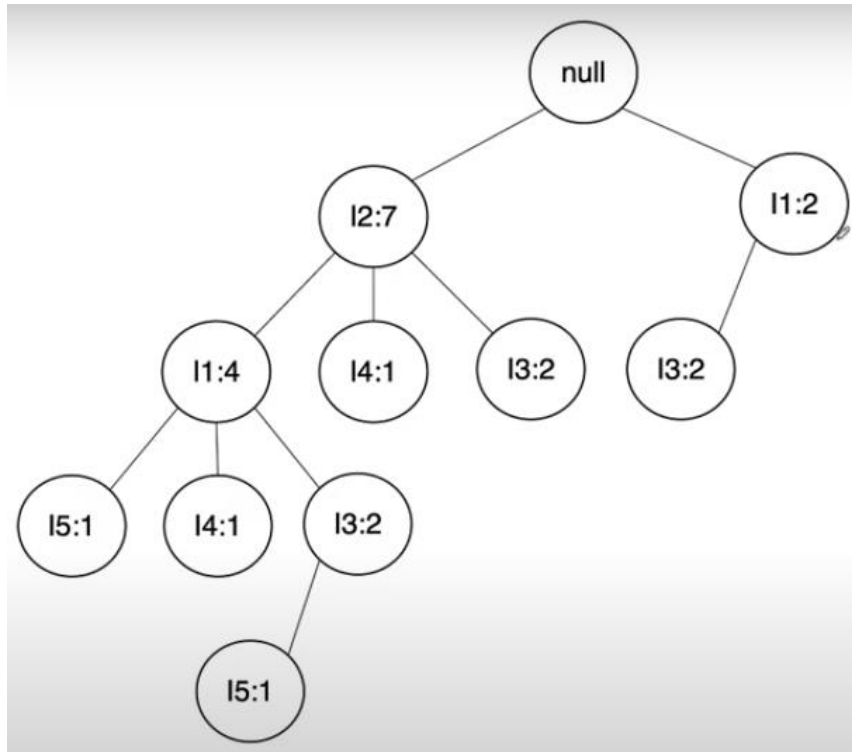


- Langkah 3 : Urutkan Tabel Transaksi berdasarkan frequent 1 item set yang sdh diurutkan dari terbesar .
I2, I1, I3, I4 dan I5

<i>TID</i>	<i>List of item_IDs</i>
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3

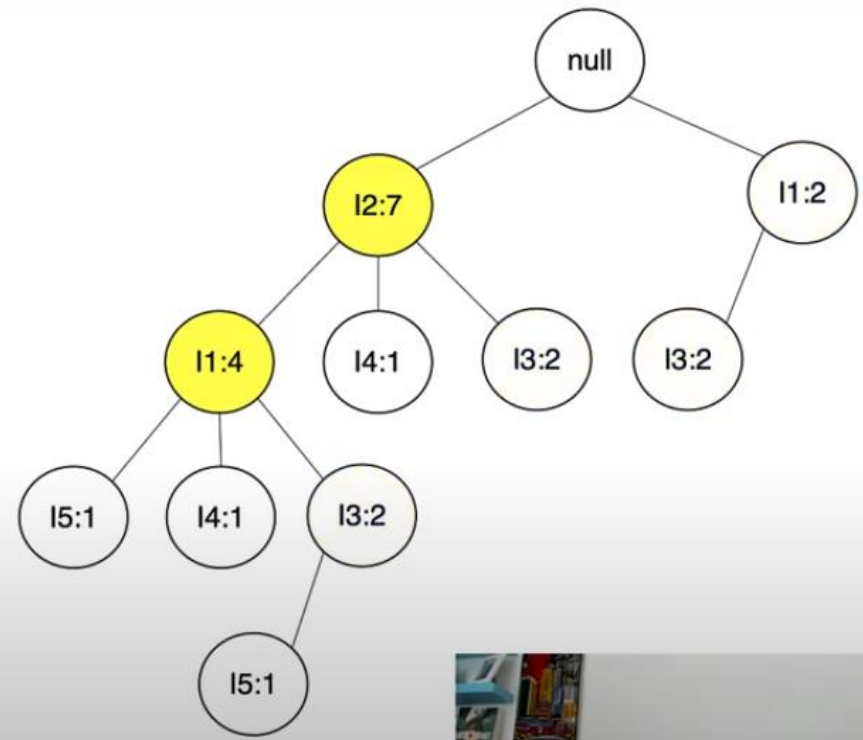
TID	List of Items
T1	I2, I1, I5
T2	I2, I4
T3	I2, I3
T4	I2, I1, I4
T5	I1, I3
T6	I2, I3
T7	I1, I3
T8	I2, I1, I3, I5
T9	I2, I1, I3

- Langkah 4 : Bentuk Fp Tree sesuai algoritma FP Tree

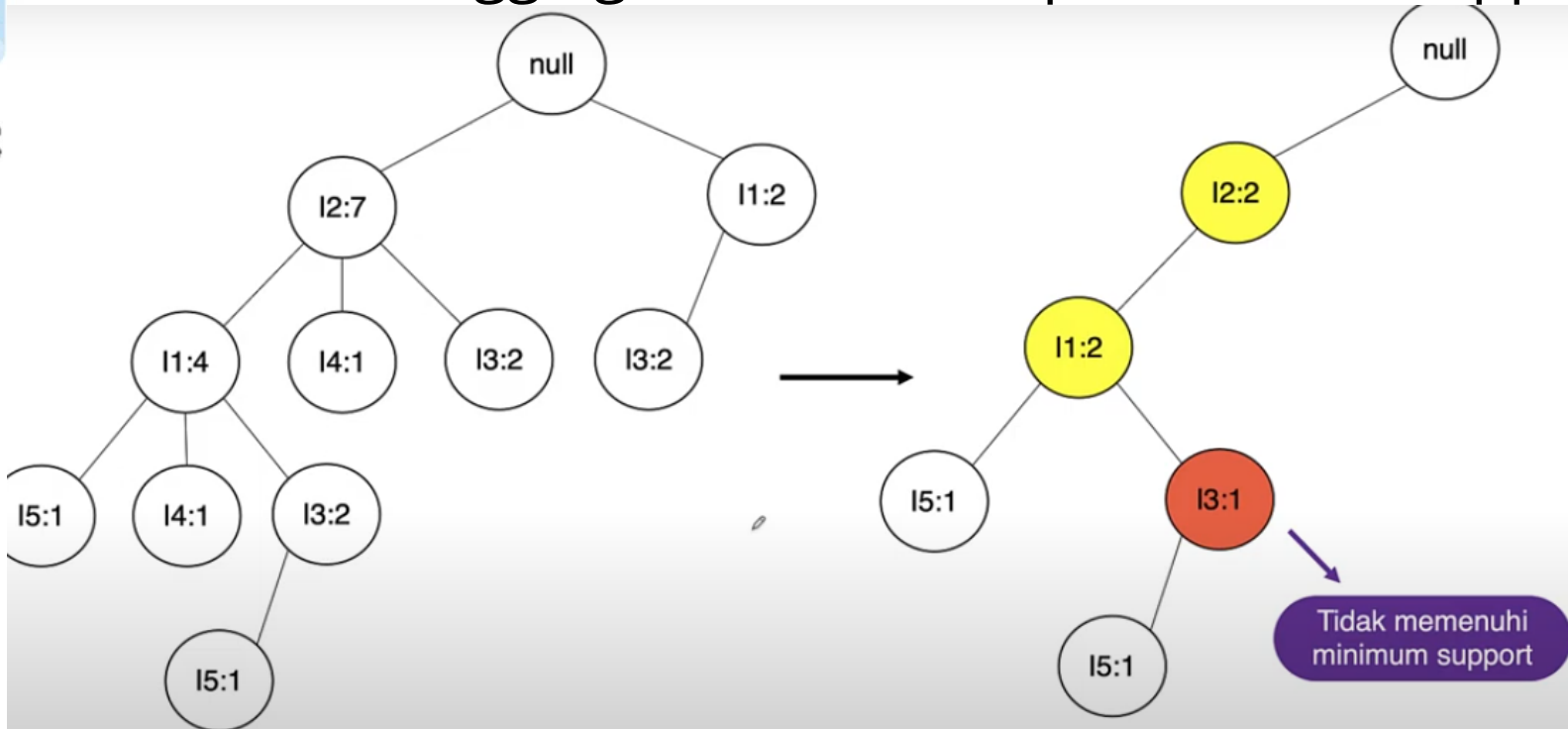


- Langkah 5 : Bentuk Conditional Pattern Base dimulai dengan support count terendah ke item dengan support count tertinggi

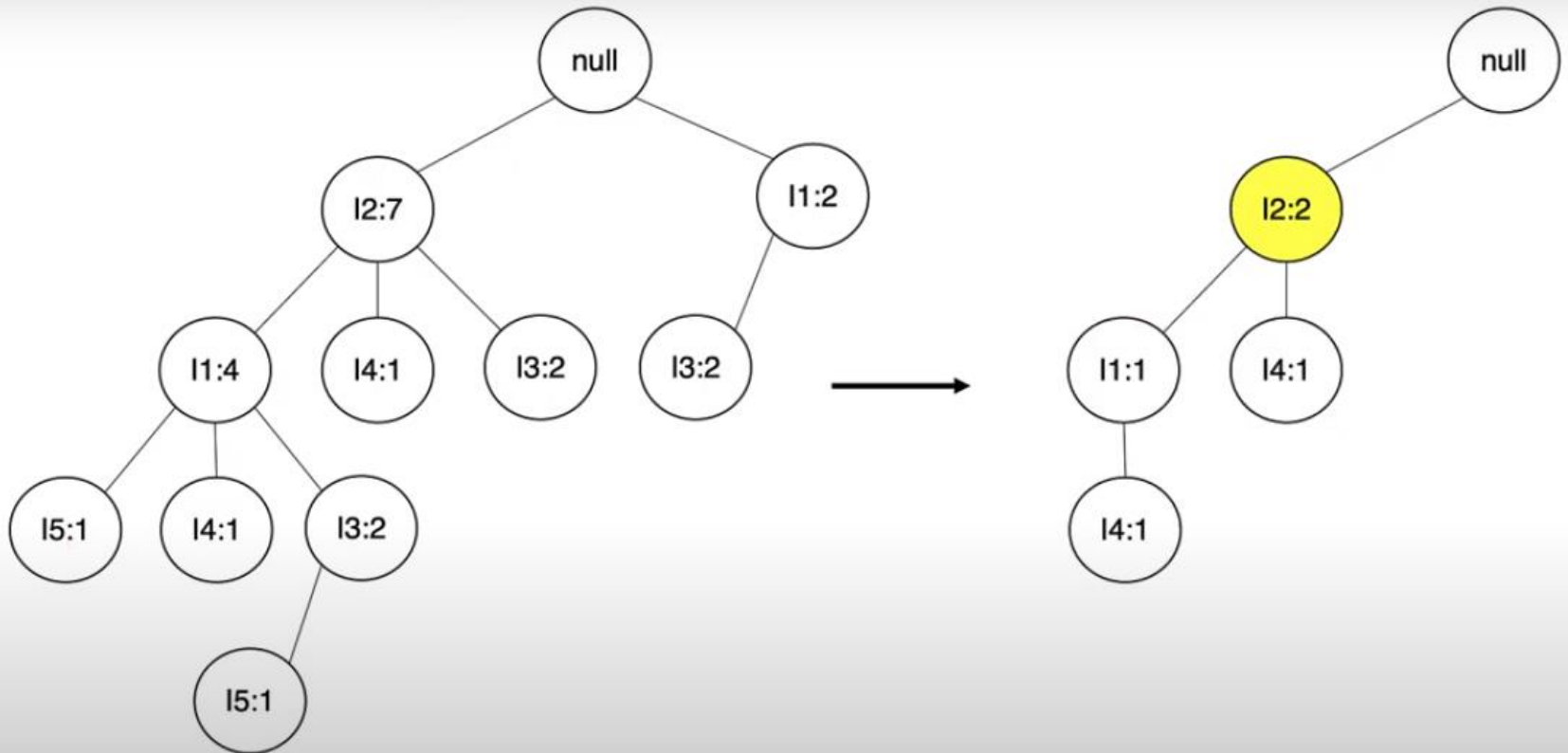
Item	Conditional Pattern Base
I5	{I2, I1:1}, {I2, I1, I3:1}
I4	{I2, I1:1}, {I2:1}
I3	{I2, I1, I3:2}, {I2:2}, {I1:2}
I1	{I2:4}



- Langkah 6 Bentuk conditional FP tree dimulai dengan support count terendah ke item dengan support count tertinggi (gunakan konsep minimum support)



Tahap 1: Conditional FP-Tree untuk I5 = {I2:2, I1:2}



Tahap 2: Conditional FP-Tree untuk $l_4 = \{l_2:2\}^o$



ITEM	CONDITIONAL PATTERN BASE	CONDITIONAL FP-TREE
I5	{I2, I1:1}, {I2, I1, I3:1}	{I2:2, I1:2}
I4	{I2, I1:1}, {I2:1}	{I2:2}
I3	{I2, I1:2}, {I2:2}, {I1:2}	{I2:4, I1:2}, {I1:2}
I1	{I2:4}	{I2:4}

- Langkah 7 : Bentuk Frequent patterns dengan cara menjoinkan set dan subset conditional FP tree dengan Item

ITEM	CONDITIONAL PATTERN BASE	CONDITIONAL FP-TREE	FREQUENT PATTERNS GENERATED
I5	{I2, I1:1}, {I2, I1, I3:1}	{I2:2, I1:2}	{I2, I5:2}, {I1, I5:2}, {I2, I1, I5:2}
I4	{I2, I1:1}, {I2:1}	{I2:2}	{I2, I4:2}
I3	{I2, I1, I3:2}, {I2:2}, {I1:2}	{I2:4, I1:2}, {I1:2}	{I2, I3:4}, {I1, I3:4}, {I2, I1, I3:2}
I1	{I2:4}	{I2:4}	{I2, I1:4}

- Break down setiap pattern
- Misalnya (I2, I1, I3:2) menjadi
{
 (I2, I1, I3),
 (I2, I1), (I2, I3), (I1, I3)
 (I2), (I1), (I3)
}

- Langkah 8 : cari Strong Asosiation Rule berdasarkan Frequent Pattern yang terbentuk dengan cara yang sama dengan apriori

$\{I1, I2, I5\} \rightarrow \text{Subset} = \{\{I1, I2\}, \{I1, I5\}, \{I2, I5\}, \{I1\}, \{I2\}, \{I5\}\}$
Minimum confidence: 70%

- IF $\{I1, I2\}$ THEN $\{I5\}$ Confidence: $sc\{I1, I2, I5\} / sc\{I1, I2\} = 2/4 = 50\%$ (Rule Rejected!)
- IF $\{I1, I5\}$ THEN $\{I2\}$ Confidence: $sc\{I1, I2, I5\} / sc\{I1, I5\} = 2/2 = 100\%$ (Rule Selected!)
- IF $\{I2, I5\}$ THEN $\{I1\}$ Confidence: $sc\{I1, I2, I5\} / sc\{I2, I5\} = 2/2 = 100\%$ (Rule Selected!)
- IF $\{I1\}$ THEN $\{I2, I5\}$ Confidence: $sc\{I1, I2, I5\} / sc\{I1\} = 2/6 = 33\%$ (Rule Rejected!)
- IF $\{I2\}$ THEN $\{I1, I5\}$ Confidence: $sc\{I1, I2, I5\} / sc\{I2\} = 2/7 = 29\%$ (Rule Rejected!)
- IF $\{I5\}$ THEN $\{I1, I2\}$ Confidence: $sc\{I1, I2, I5\} / sc\{I5\} = 2/2 = 100\%$ (Rule Selected!)

- Langkah 9 : Representasi Pengetahuan, berdasarkan tujuan data mining

No.	Strong Association Rule	Representasi Pengetahuan
1	if {I1,I5} then {I2}	Item I1, I2, dan I5 harus ditempatkan pada rak yang berdekatan/pada satu lorong rak
2	if {I3} then {I1,I2}	Item I3, I1, dan I2 harus ditempatkan pada rak yang berdekatan/pada satu lorong rak



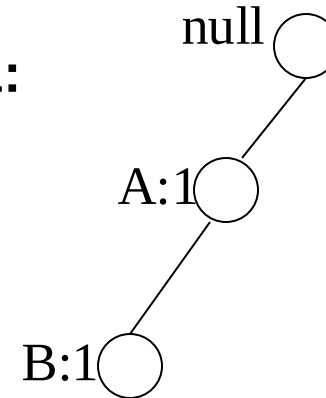
FP-Growth Algorithm

- Use a compressed representation of the database using an **FP-tree**.
- Once an FP-tree has been constructed, it uses a recursive divide-and-conquer approach to mine the frequent itemsets.

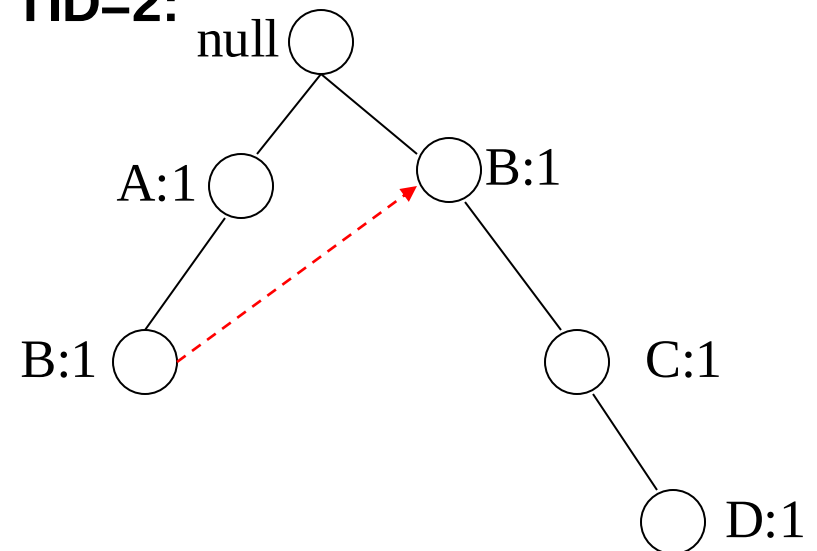
FP-Tree Construction

TID	Items
1	{A,B}
2	{B,C,D}
3	{A,C,D,E}
4	{A,D,E}
5	{A,B,C}
6	{A,B,C,D}
7	{B,C}
8	{A,B,C}
9	{A,B,D}
10	{B,C,E}

After reading TID=1:



After reading TID=2:



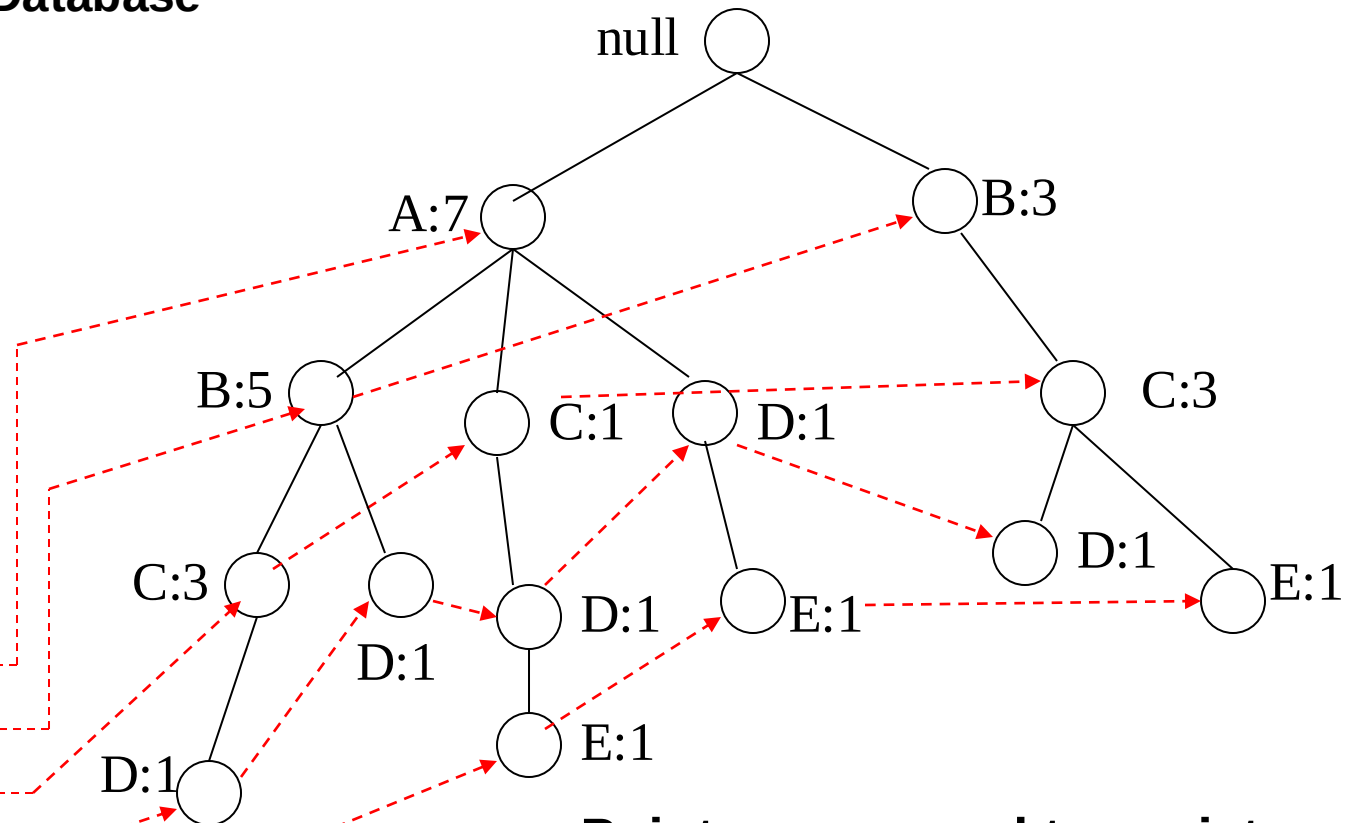
FP-Tree Construction (cont.)

TID	Items
1	{A,B}
2	{B,C,D}
3	{A,C,D,E}
4	{A,D,E}
5	{A,B,C}
6	{A,B,C,D}
7	{B,C}
8	{A,B,C}
9	{A,B,D}
10	{B,C,E}

Transaction Database

Header table

Item	Pointer
A	
B	
C	
D	
E	



Pointers are used to assist frequent itemset generation

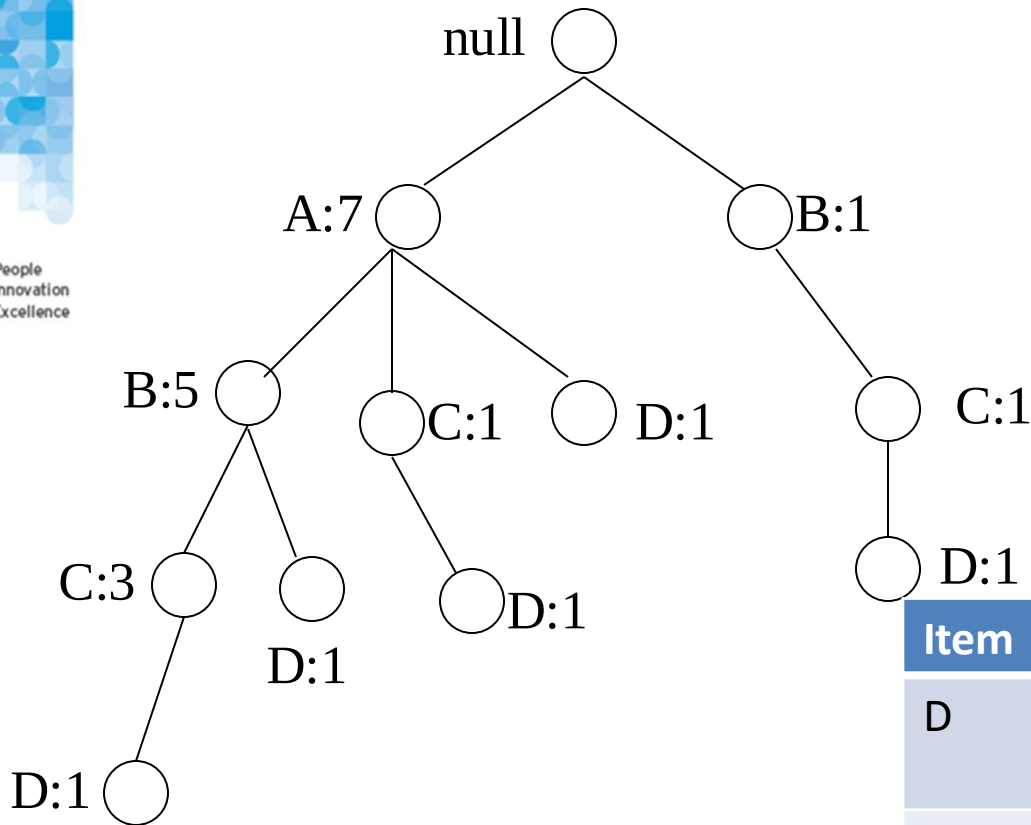
FP-Growth

Conditional Pattern base for D:

$$P = \{(A:1, B:1, C:1), \\ (A:1, B:1), \\ (A:1, C:1), \\ (A:1), \\ (B:1, C:1)\}$$

Recursively apply FP-growth on P

Frequent Itemsets found
(with sup > 1):
AD, BD, CD, ACD, BCD



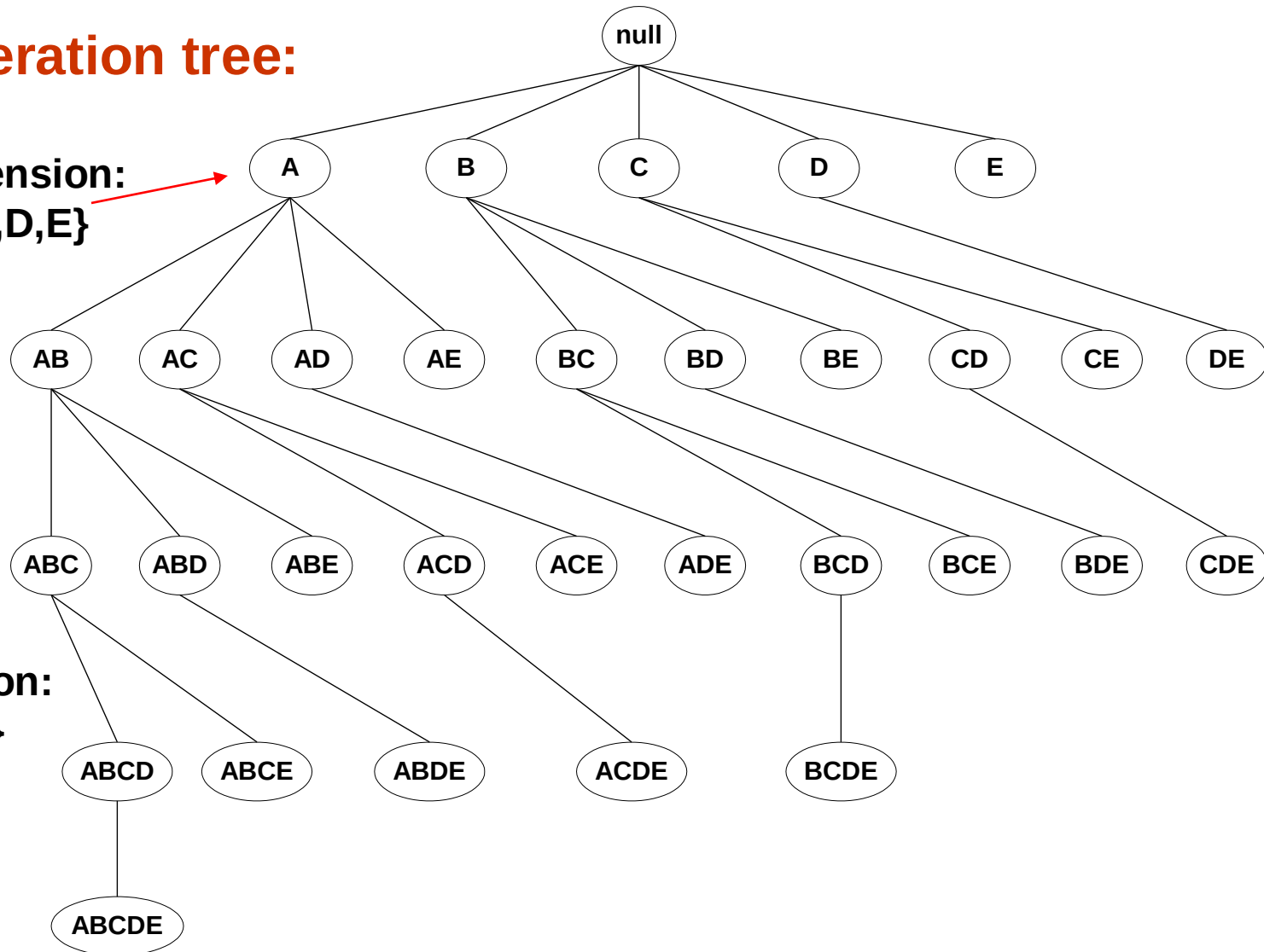
Item	Conditional Pattern
D	{A, B, C :1 ; A, B:1 ; A, C :1; A:1 ; B, C: 1}

Tree Projection

Set enumeration tree:

Possible Extension:

$E(A) = \{B, C, D, E\}$



Possible Extension:

$E(ABC) = \{D, E\}$

Why is FP-Growth the Winner?

- Divide-and-conquer:
 - decompose both the mining task and DB according to the frequent patterns obtained so far
 - leads to focused search of smaller databases
- Other factors
 - no candidate generation, no candidate test
 - compressed database: FP-tree structure
 - no repeated scan of entire database
 - basic ops—counting local freq items and building sub FP-tree, **no pattern search and matching**

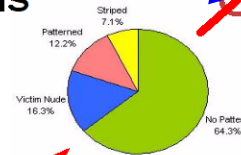
Application of Interestingness Measure

Interestingness Measures

Knowledge

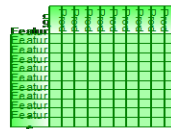


Patterns



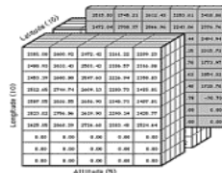
Postprocessing

Preprocessed Data



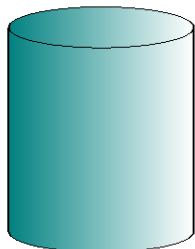
Mining

Selected Data



Preprocessing

Data



Selection

Subjective Interestingness Measure

- **Objective measure:**
 - Rank patterns based on statistics computed from data.
 - e.g., 21 measures of association (support, confidence, Laplace, Gini, mutual information, Jaccard, etc).
- **Subjective measure:**
 - Rank patterns according to user's interpretation
 - A pattern is subjectively interesting if it contradicts the expectation of a user (Silberschatz & Tuzhilin).
 - A pattern is subjectively interesting if it is actionable (Silberschatz & Tuzhilin).

Exercise

Transaction ID	Items
T1	HotDogs, Buns, Ketchup
T2	HotDogs, Buns
T3	HotDogs, Coke, Chips
T4	Chips, Coke
T5	Chips, Ketchup
T6	HotDogs, Coke, Chips

Trace the results of using the Apriori algorithm on the grocery store transactions above with support threshold $s=33.34\%$ and confidence threshold $c=60\%$. Show the candidate and frequent itemsets for each database scan. Enumerate all the final frequent itemsets. Also indicate the association rules that are generated and highlight the strong ones, sort them by confidence.

References

1. Bishop, C. M. (2006). *Pattern recognition and machine learning*. springer.
2. Harrington, P. (2012). *Machine learning in action*. Manning Publications Co..

Q&A

Thank You